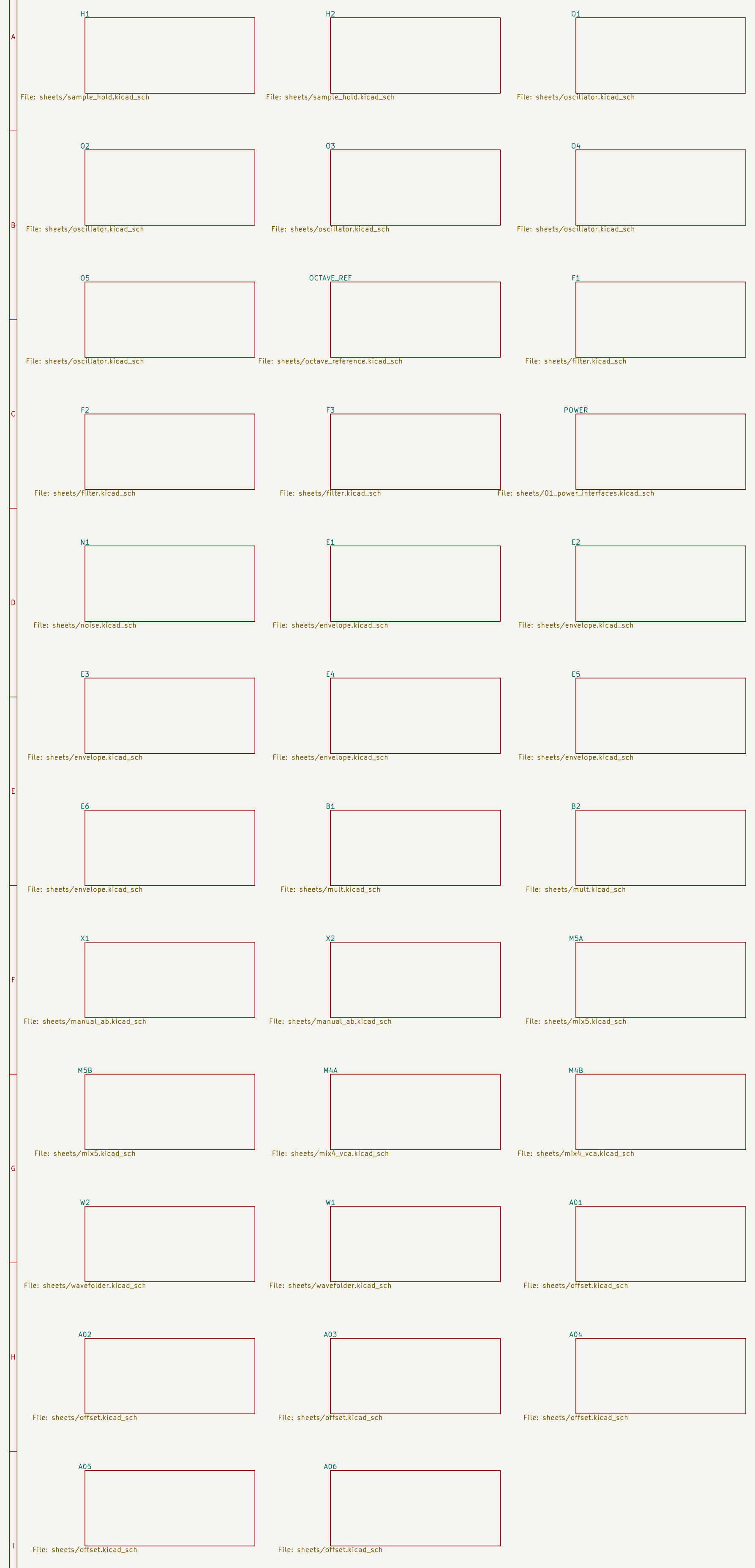
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H2
File: sheets/sample_hold.kicad_sch
O1
File: sheets/oscillator.kicad_sch
O2
File: sheets/oscillator.kicad_sch
O3
File: sheets/oscillator.kicad_sch
O4
File: sheets/oscillator.kicad_sch
O5
File: sheets/oscillator.kicad_sch
OCTAVE_REF
File: sheets/octave_reference.kicad_sch
F1
File: sheets/filter.kicad_sch
F2
File: sheets/filter.kicad_sch
F3
File: sheets/filter.kicad_sch
POWER
File: sheets/01_power_interfaces.kicad_sch
N1
File: sheets/noise.kicad_sch
E1
File: sheets/envelope.kicad_sch
E2
File: sheets/envelope.kicad_sch
E3
File: sheets/envelope.kicad_sch
E4
File: sheets/envelope.kicad_sch
E5
File: sheets/envelope.kicad_sch
E6
File: sheets/envelope.kicad_sch
B1
File: sheets/mult.kicad_sch
B2
File: sheets/mult.kicad_sch
X1
File: sheets/manual_ab.kicad_sch
X2
File: sheets/manual_ab.kicad_sch
M5A
File: sheets/mix5.kicad_sch
M5B
File: sheets/mix5.kicad_sch
M4A
File: sheets/mix4_vca.kicad_sch
M4B
File: sheets/mix4_vca.kicad_sch
W2
File: sheets/wavefolder.kicad_sch
W1
File: sheets/wavefolder.kicad_sch
A01
File: sheets/offset.kicad_sch
A02
File: sheets/offset.kicad_sch
A03
File: sheets/offset.kicad_sch
A04
File: sheets/offset.kicad_sch
A05
File: sheets/offset.kicad_sch
A06
File: sheets/offset.kicad_sch

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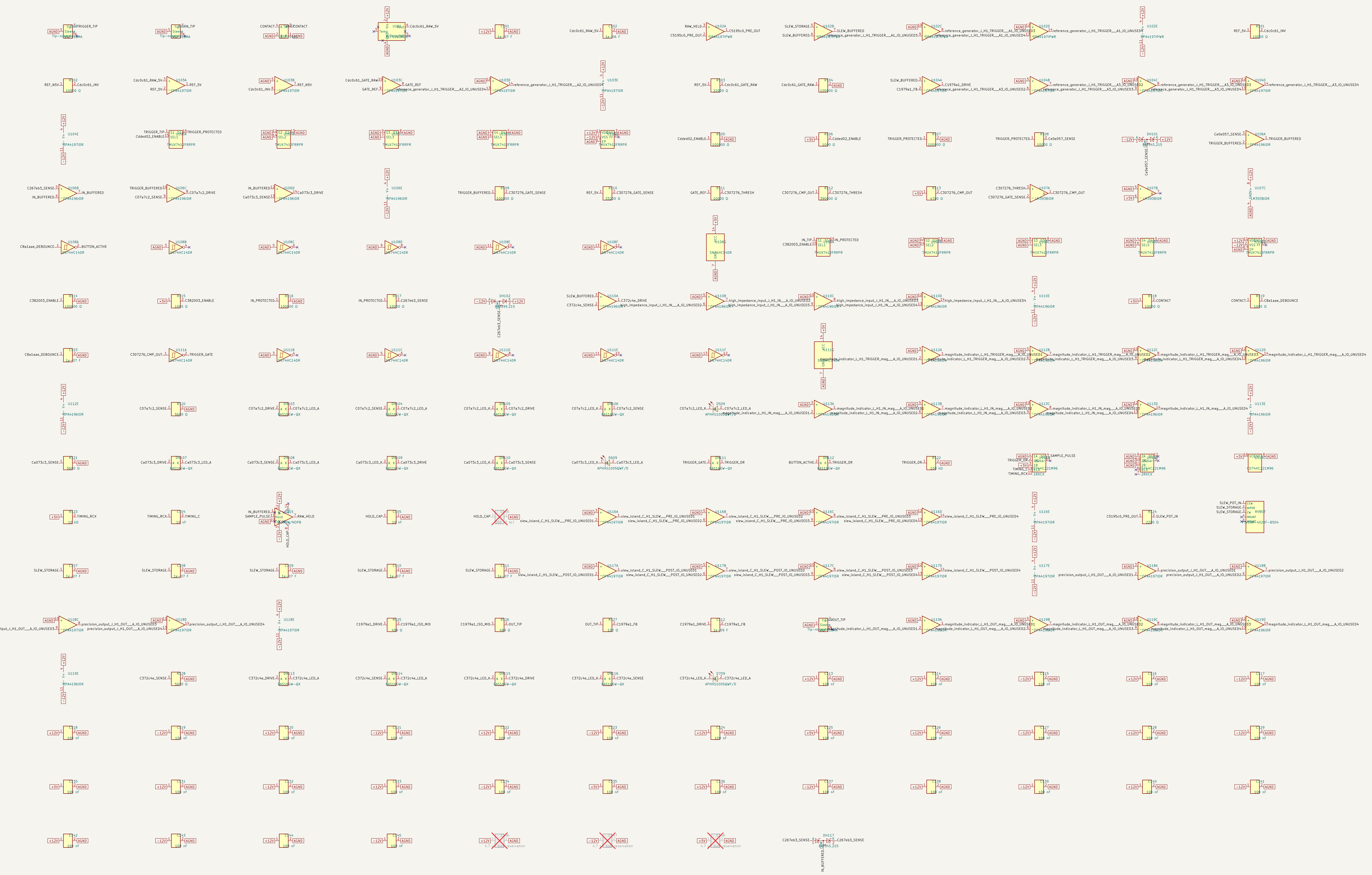
| Size: A0 | Date: |
| | **Rev:** |
| KiCad E.D.A. 10.0.6 | Id: 1/36 |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DH102
DH117
D509
D609
D709
RV807
OPA4197IPWR
OPA4197IDR
OPA4196IDR
LM393BIDR
SN74HC14DR
TMUX7412FRRPR
CD74HC221M96
BAS16W-QX
APHHS1005QWF/D
BAT54S,215
AGND
+12V
-12V
+5V
TRIGGER_TIP
CONTACT
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Cdc0c61_INV
Cdc0c61_GATE_RAW
REF_N5V
REF_5V
GATE_REF
RAW_HELD
C5195c0_PRE_OUT
SLEW_STORAGE
SLEW_BUFFERED
C1979a1_FB
C1979a1_DRIVE
C1979a1_ISO_MID
TRIGGER_PROTECTED
TRIGGER_BUFFERED
Cdded02_ENABLE
Ce0e057_SENSE
C267eb3_SENSE
IN_BUFFERED
C07a7c2_SENSE
C07a7c2_DRIVE
C07a7c2_LED_A
C07a7c2_LED_K
Ca073c3_SENSE
Ca073c3_DRIVE
Ca073c3_LED_A
Ca073c3_LED_K
C307276_GATE_SENSE
C307276_THRESH
C307276_CMP_OUT
C8a1aae_DEBOUNCE
BUTTON_ACTIVE
C382003_ENABLE
IN_PROTECTED
IN_TIP
C372c4e_DRIVE
C372c4e_SENSE
C372c4e_LED_A
C372c4e_LED_K
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TRIGGER_OR
SAMPLE_PULSE
TIMING_RCX
TIMING_C
HOLD_CAP
SLEW_POT_IN
OUT_TIP
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magnitude_indicator_J_H1_IN_mag___A_IO_UNUSED1
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slew_island_C_H1_SLEW___POST_IO_UNUSED1
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magnitude_indicator_J_H1_OUT_mag___A_IO_UNUSED1
DNP
Reservation
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H
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J
K

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U202C
U202D
U202E
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OPA4197IDR
OPA4196IDR
SN74HC14DR
TMUX7412FRRPR
LM393BIDR
BAS16GW-QX
APHHS1005QWF/D
BAT54S,215
TRIGGER_TIP
IN_TIP
OUT_TIP
CONTACT
RAW_HELD
SLEW_STORAGE
SLEW_BUFFERED
TRIGGER_PROTECTED
TRIGGER_BUFFERED
IN_BUFFERED
IN_PROTECTED
TRIGGER_GATE
TRIGGER_OR
BUTTON_ACTIVE
SAMPLE_PULSE
TIMING_RCX
TIMING_C
HOLD_CAP
REF_5V
REF_N5V
GATE_REF
SLEW_POT_IN
+12V
-12V
+5V
AGND

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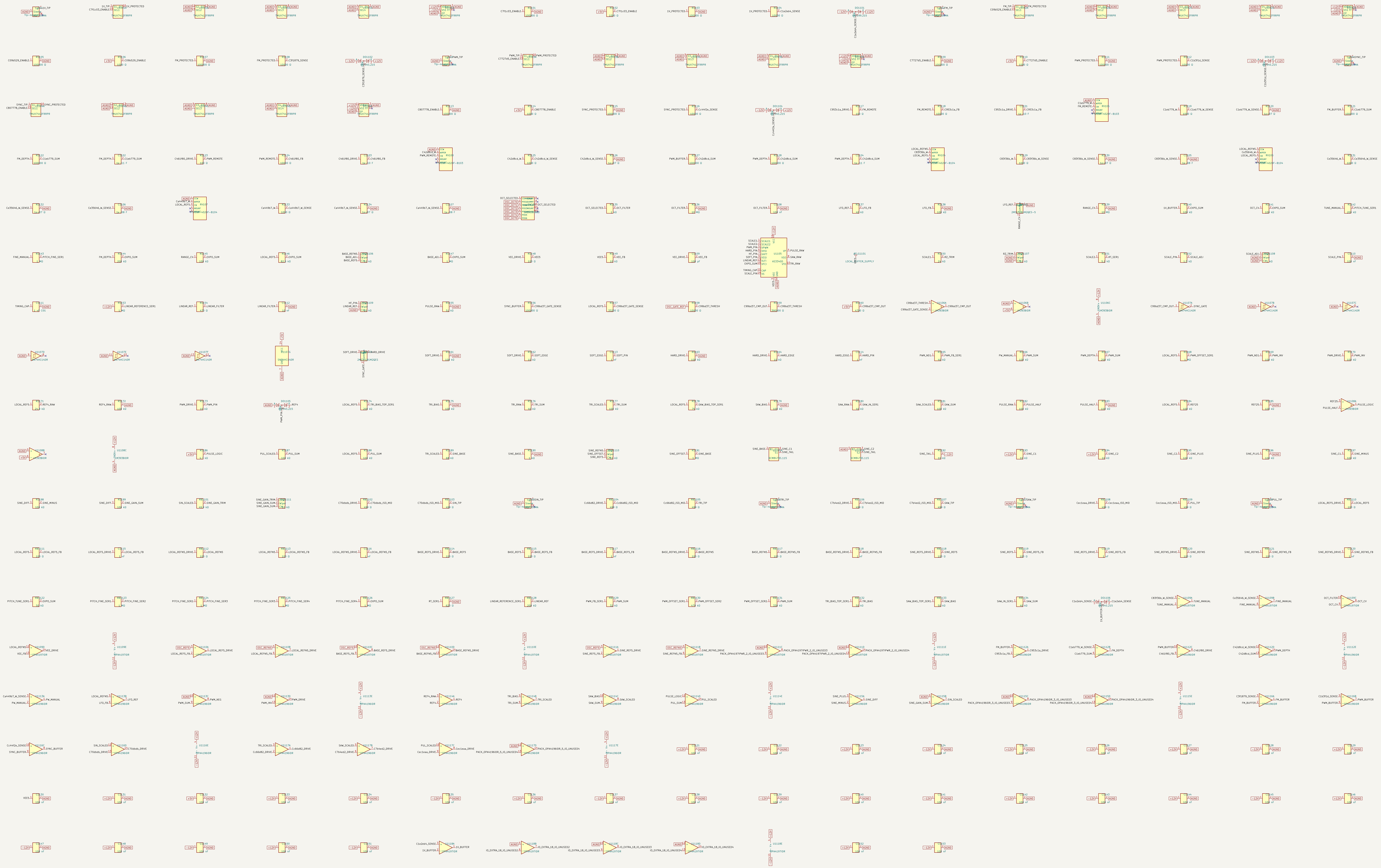
Id: 3/36



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KiCad E.D.A. 10.0.6 Id: 4/36

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Size: A0 | Date:
**Rev:**
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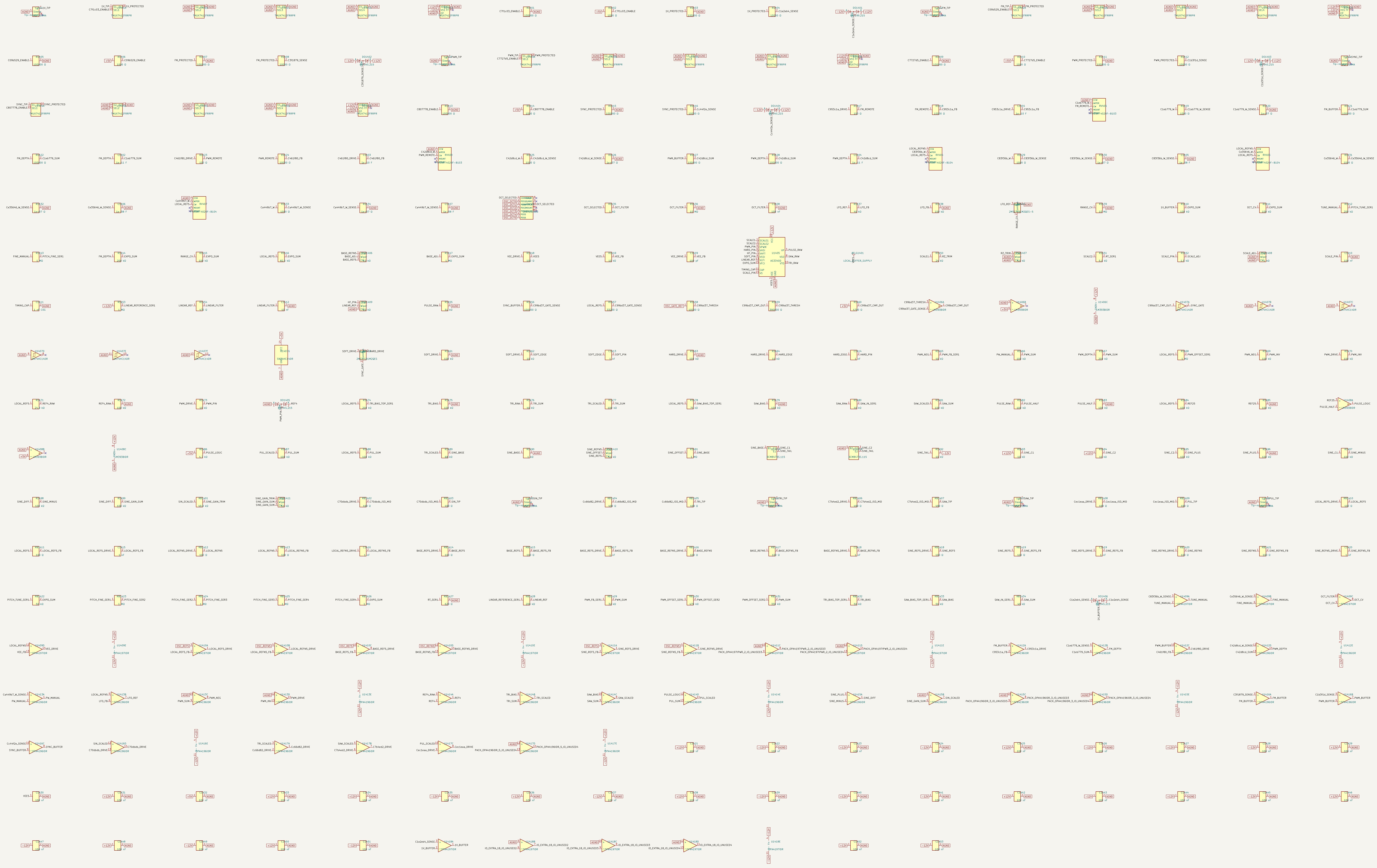
Sheet: /03/
File: oscillator.kicad_sch

**Title:**

Size: A0
Date:
**Rev:**
KiCad E.D.A. 10.0.6
Id: 6/36



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KiCad E.D.A. 10.0.6 Id: 7/36

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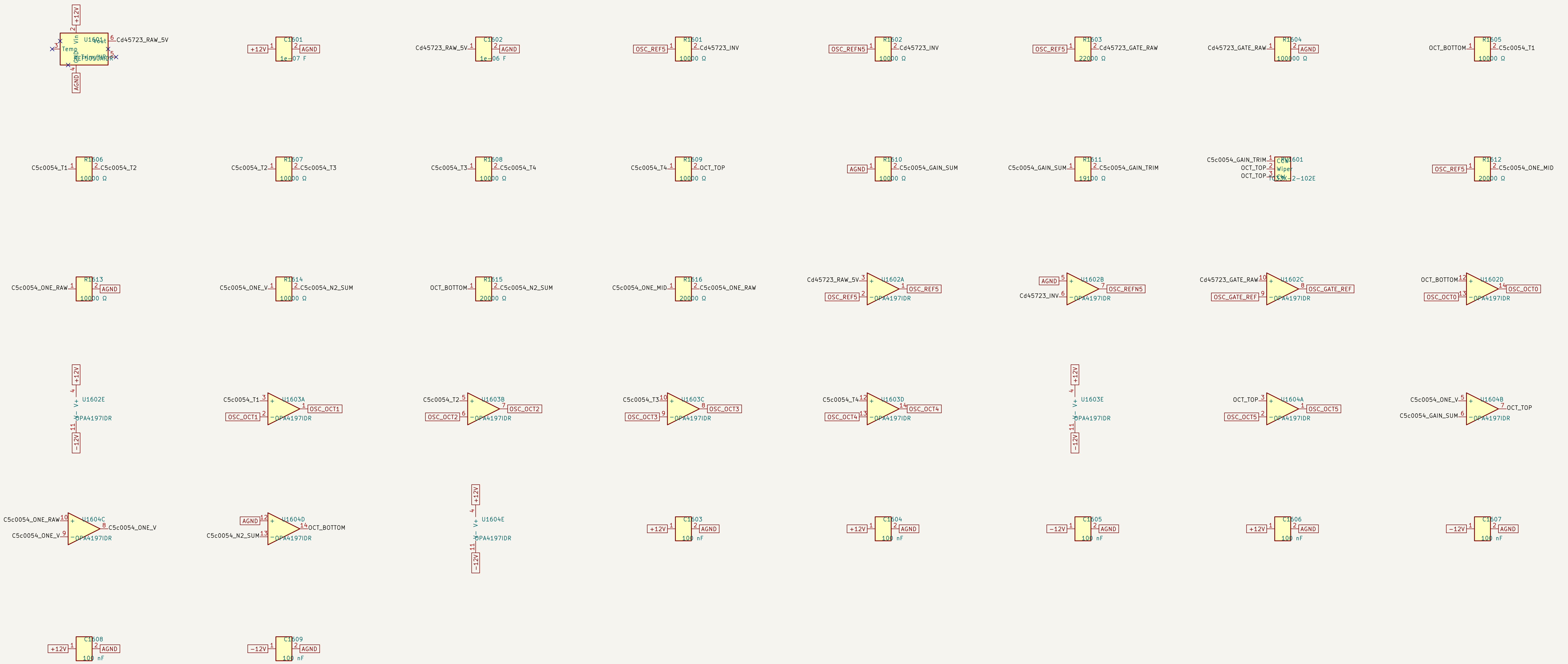
Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 8/36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Vin
Vout
Temp
GND
Trim/NR
REF5050AIDR
+12V
AGND
Cd45723_RAW_5V
C1601 +12V AGND 1e-07 F
C1602 Cd45723_RAW_5V AGND 1e-06 F
R1601 OSC_REF5 Cd45723_INV 10000 Ω
R1602 OSC_REFN5 Cd45723_INV 10000 Ω
R1603 OSC_REF5 Cd45723_GATE_RAW 22000 Ω
R1604 Cd45723_GATE_RAW AGND 100000 Ω
R1605 OCT_BOTTOM C5c0054_T1 10000 Ω
R1606 C5c0054_T1 C5c0054_T2 10000 Ω
R1607 C5c0054_T2 C5c0054_T3 10000 Ω
R1608 C5c0054_T3 C5c0054_T4 10000 Ω
R1609 C5c0054_T4 OCT_TOP 10000 Ω
R1610 AGND C5c0054_GAIN_SUM 10000 Ω
R1611 C5c0054_GAIN_SUM C5c0054_GAIN_TRIM 19100 Ω
RV1601 C5c0054_GAIN_TRIM OCT_TOP OCT_TOP CCW Wiper CW TC33X-2-102E
R1612 OSC_REF5 C5c0054_ONE_MID 20000 Ω
R1613 C5c0054_ONE_RAW AGND 10000 Ω
R1614 C5c0054_ONE_V C5c0054_N2_SUM 10000 Ω
R1615 OCT_BOTTOM C5c0054_N2_SUM 20000 Ω
R1616 C5c0054_ONE_MID C5c0054_ONE_RAW 20000 Ω
U1602A Cd45723_RAW_5V OSC_REF5 OSC_REF5 OPA4197IDR
U1602B AGND Cd45723_INV OSC_REFN5 OPA4197IDR
U1602C Cd45723_GATE_RAW OSC_GATE_REF OSC_GATE_REF OPA4197IDR
U1602D OCT_BOTTOM OSC_OCT0 OSC_OCT0 OPA4197IDR
U1602E +12V −12V V+ V− OPA4197IDR
U1603A C5c0054_T1 OSC_OCT1 OSC_OCT1 OPA4197IDR
U1603B C5c0054_T2 OSC_OCT2 OSC_OCT2 OPA4197IDR
U1603C C5c0054_T3 OSC_OCT3 OSC_OCT3 OPA4197IDR
U1603D C5c0054_T4 OSC_OCT4 OSC_OCT4 OPA4197IDR
U1603E +12V −12V V+ V− OPA4197IDR
U1604A OCT_TOP OSC_OCT5 OSC_OCT5 OPA4197IDR
U1604B C5c0054_ONE_V C5c0054_GAIN_SUM OCT_TOP OPA4197IDR
U1604C C5c0054_ONE_RAW C5c0054_ONE_V C5c0054_ONE_V OPA4197IDR
U1604D AGND C5c0054_N2_SUM OCT_BOTTOM OPA4197IDR
U1604E +12V −12V V+ V− OPA4197IDR
C1603 +12V AGND 100 nF
C1604 +12V AGND 100 nF
C1605 −12V AGND 100 nF
C1606 +12V AGND 100 nF
C1607 −12V AGND 100 nF
C1608 +12V AGND 100 nF
C1609 −12V AGND 100 nF

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Date:
**Rev:**
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Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 10/36

Sheet: /F2/
File: filter.kicad_sch

**Title:**

Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 11/36

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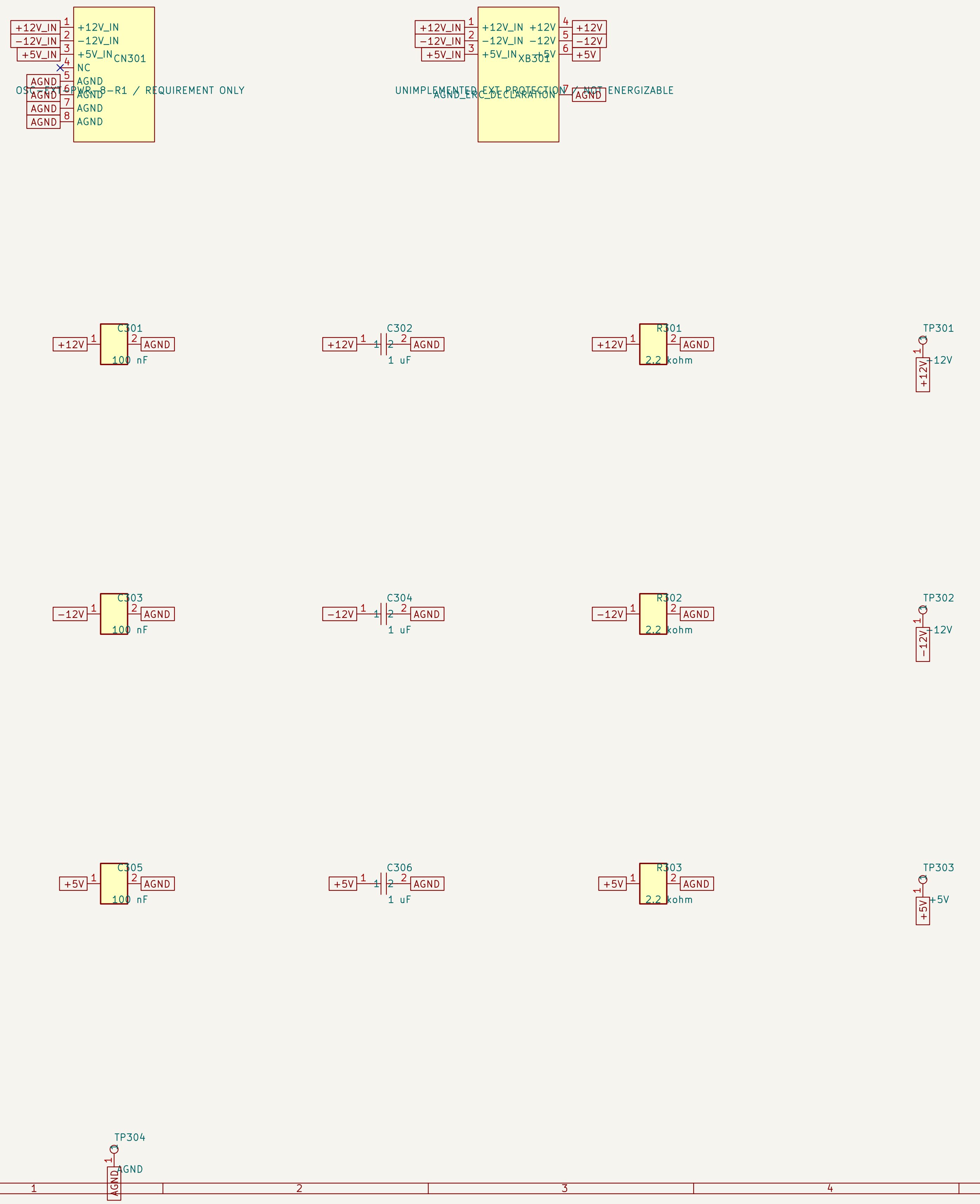
KiCad E.D.A. 10.0.6 | Id: 12/36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OSC-EXT-PWR-8-R1 / REQUIREMENT ONLY
+12V_IN
−12V_IN
+5V_IN
NC
AGND
AGND
AGND
AGND
1
2
3
4
5
6
7
8
XB301
UNIMPLEMENTED EXT PROTECTION / NOT ENERGIZABLE
+12V_IN
−12V_IN
+5V_IN
+12V
−12V
+5V
AGND_ERC_DECLARATION
AGND
C301
100 nF
C302
1 uF
R301
2.2 kohm
TP301
+12V
C303
100 nF
C304
1 uF
R302
2.2 kohm
TP302
−12V
C305
100 nF
C306
1 uF
R303
2.2 kohm
TP303
+5V
TP304
AGND

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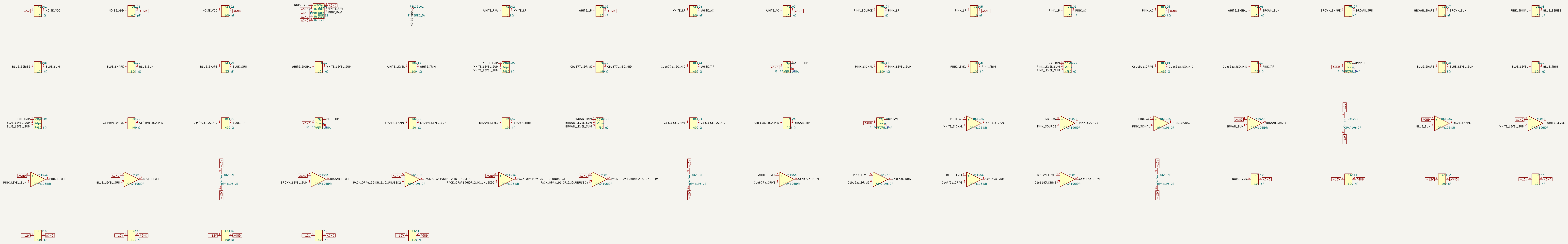
Date:
**Rev:**
KiCad E.D.A. 10.0.6
Id: 13/36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C6101 NOISE_VDD 1 2 AGND 4.7 µF
C6102 NOISE_VDD 1 2 AGND 100 nF
U6101 NOISE_VDD 1 +5V0V 8 AGND
AGND 2 WHITE Out Unused 3 WHITE_RAW
AGND 4 PINK Out Unused 7 PINK_RAW
AGND 5 Unused NOISE2
AGND 6 Unused
#FLG6101 NOISE_VDD FILTERED_5V
R6102 WHITE_RAW 1 2 WHITE_LP 1 kΩ
C6103 WHITE_LP 1 2 AGND 10 nF
C6104 WHITE_LP 1 2 WHITE_AC 100 nF
R6103 WHITE_AC 1 2 AGND 100 kΩ
R6104 PINK_SOURCE 1 2 PINK_LP 1 kΩ
C6105 PINK_LP 1 2 AGND 10 nF
C6106 PINK_LP 1 2 PINK_AC 100 nF
R6105 PINK_AC 1 2 AGND 100 kΩ
R6106 WHITE_SIGNAL 1 2 BROWN_SUM 100 kΩ
R6107 BROWN_SHAPE 1 2 BROWN_SUM 1 MΩ
C6107 BROWN_SHAPE 1 2 BROWN_SUM 10 nF
C6108 PINK_SIGNAL 1 2 BLUE_SERIES 100 pF
R6108 BLUE_SERIES 1 2 BLUE_SUM 100 kΩ
R6109 BLUE_SHAPE 1 2 BLUE_SUM 100 kΩ
C6109 BLUE_SHAPE 1 2 BLUE_SUM 33 pF
R6110 WHITE_SIGNAL 1 2 WHITE_LEVEL_SUM 100 kΩ
R6111 WHITE_LEVEL 1 2 WHITE_TRIM 100 kΩ
RV6101 WHITE_TRIM 1 CCW WHITE_LEVEL_SUM 2 Wiper WHITE_LEVEL_SUM 3 CW 10 kΩ
R6112 Cbe877b_DRIVE 1 2 Cbe877b_ISO_MID 499 Ω
R6113 Cbe877b_ISO_MID 1 2 WHITE_TIP 499 Ω
AGND S Sleeve Tip T WHITE_TIP TN Tip-normal WQP518MA
R6114 PINK_SIGNAL 1 2 PINK_LEVEL_SUM 100 kΩ
R6115 PINK_LEVEL 1 2 PINK_TRIM 100 kΩ
RV6102 PINK_TRIM 1 CCW PINK_LEVEL_SUM 2 Wiper PINK_LEVEL_SUM 3 CW 10 kΩ
R6116 Cdbc5aa_DRIVE 1 2 Cdbc5aa_ISO_MID 499 Ω
R6117 Cdbc5aa_ISO_MID 1 2 PINK_TIP 499 Ω
AGND S Sleeve Tip T PINK_TIP TN Tip-normal WQP518MA
R6118 BLUE_SHAPE 1 2 BLUE_LEVEL_SUM 10 kΩ
R6119 BLUE_LEVEL 1 2 BLUE_TRIM 100 kΩ
RV6103 BLUE_TRIM 1 CCW BLUE_LEVEL_SUM 2 Wiper BLUE_LEVEL_SUM 3 CW 10 kΩ
R6120 Ce44f9a_DRIVE 1 2 Ce44f9a_ISO_MID 499 Ω
R6121 Ce44f9a_ISO_MID 1 2 BLUE_TIP 499 Ω
AGND S Sleeve Tip T BLUE_TIP TN Tip-normal WQP518MA
R6122 BROWN_SHAPE 1 2 BROWN_LEVEL_SUM 20 kΩ
R6123 BROWN_LEVEL 1 2 BROWN_TRIM 100 kΩ
RV6104 BROWN_TRIM 1 CCW BROWN_LEVEL_SUM 2 Wiper BROWN_LEVEL_SUM 3 CW 10 kΩ
R6124 Cde1183_DRIVE 1 2 Cde1183_ISO_MID 499 Ω
R6125 Cde1183_ISO_MID 1 2 BROWN_TIP 499 Ω
AGND S Sleeve Tip T BROWN_TIP TN Tip-normal WQP518MA
U6102A WHITE_AC 3 + WHITE_SIGNAL 2 − 1 WHITE_SIGNAL OPA4196IDR
U6102B PINK_RAW 5 + PINK_SOURCE 6 − 7 PINK_SOURCE OPA4196IDR
U6102C PINK_AC 10 + PINK_SIGNAL 9 − 8 PINK_SIGNAL OPA4196IDR
U6102D AGND 12 + BROWN_SUM 13 − 14 BROWN_SHAPE OPA4196IDR
U6102E +12V 4 V+ V− 11 −12V OPA4196IDR
U6103A AGND 3 + BLUE_SUM 2 − 1 BLUE_SHAPE OPA4196IDR
U6103B AGND 5 + WHITE_LEVEL_SUM 6 − 7 WHITE_LEVEL OPA4196IDR
U6103C AGND 10 + PINK_LEVEL_SUM 9 − 8 PINK_LEVEL OPA4196IDR
U6103D AGND 12 + BLUE_LEVEL_SUM 13 − 14 BLUE_LEVEL OPA4196IDR
U6103E +12V 4 V+ V− 11 −12V OPA4196IDR
U6104A AGND 3 + BROWN_LEVEL_SUM 2 − 1 BROWN_LEVEL OPA4196IDR
U6104B AGND 5 + PACK_OPA4196IDR_2_IO_UNUSED2 6 − 7 PACK_OPA4196IDR_2_IO_UNUSED2 OPA4196IDR
U6104C AGND 10 + PACK_OPA4196IDR_2_IO_UNUSED3 9 − 8 PACK_OPA4196IDR_2_IO_UNUSED3 OPA4196IDR
U6104D AGND 12 + PACK_OPA4196IDR_2_IO_UNUSED4 13 − 14 PACK_OPA4196IDR_2_IO_UNUSED4 OPA4196IDR
U6104E +12V 4 V+ V− 11 −12V OPA4196IDR
U6105A WHITE_LEVEL 3 + Cbe877b_DRIVE 2 − 1 Cbe877b_DRIVE OPA4196IDR
U6105B PINK_LEVEL 5 + Cdbc5aa_DRIVE 6 − 7 Cdbc5aa_DRIVE OPA4196IDR
U6105C BLUE_LEVEL 10 + Ce44f9a_DRIVE 9 − 8 Ce44f9a_DRIVE OPA4196IDR
U6105D BROWN_LEVEL 12 + Cde1183_DRIVE 13 − 14 Cde1183_DRIVE OPA4196IDR
U6105E +12V 4 V+ V− 11 −12V OPA4196IDR
C6110 NOISE_VDD 1 2 AGND 100 nF
C6111 +12V 1 2 AGND 100 nF
C6112 −12V 1 2 AGND 100 nF
C6113 +12V 1 2 AGND 100 nF
C6114 −12V 1 2 AGND 100 nF
C6115 +12V 1 2 AGND 100 nF
C6116 −12V 1 2 AGND 100 nF
C6117 +12V 1 2 AGND 100 nF
C6118 −12V 1 2 AGND 100 nF

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Size: A0 Date:
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KiCad E.D.A. 10.0.6
Id: 15/36

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Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 16/36

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Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 17/36

Sheet: /E4/
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**Title:**

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 18/36

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Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 19/36

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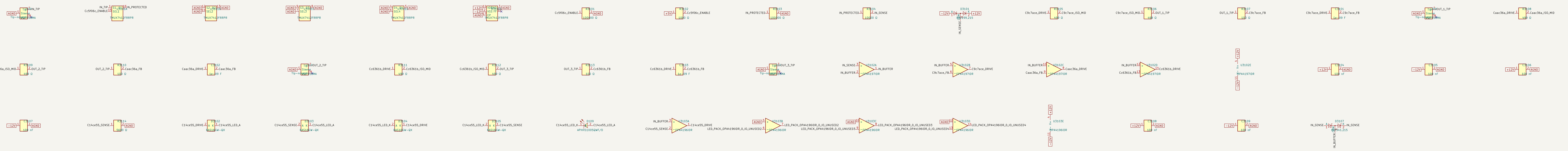
**Rev:**
KiCad E.D.A. 10.0.6
Id: 20/36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Sleeve
IN_TIP
Tip-normal
WQP518MA
IN_TIP
Cc5f06c_ENABLE
S1 D1
SEL1
U3101A
IN_PROTECTED
TMUX7412FRRPR
AGND
S2 D2
SEL2
U3101B
TMUX7412FRRPR
S3 D3
SEL3
U3101C
TMUX7412FRRPR
S4 D4
SEL4
U3101D
TMUX7412FRRPR
+12V
-12V
VDD GND
VSS FF
EP
U3101E
TMUX7412FRRPR
R3101
Cc5f06c_ENABLE
100000 Ω
R3102
+5V
1000 Ω
R3103
IN_PROTECTED
100000 Ω
R3104
IN_SENSE
10000 Ω
D3101
BAV199,215
R3105
C9c7ace_DRIVE
C9c7ace_ISO_MID
499 Ω
R3106
OUT_1_TIP
499 Ω
R3107
C9c7ace_FB
100 Ω
C3101
1e-09 F
R3108
Caac36a_DRIVE
Caac36a_ISO_MID
499 Ω
R3109
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R3110
Caac36a_FB
100 Ω
C3102
1e-09 F
R3111
Cc6361b_DRIVE
Cc6361b_ISO_MID
499 Ω
R3112
OUT_3_TIP
499 Ω
R3113
Cc6361b_FB
100 Ω
C3103
1e-09 F
U3102A
IN_BUFFER
OPA4197IDR
U3102B
U3102C
U3102D
U3102E
V+
V-
C3104
100 nF
C3105
100 nF
C3106
100 nF
C3107
100 nF
R3114
C14ce55_SENSE
5600 Ω
D3102
C14ce55_DRIVE
C14ce55_LED_A
BAS16GW-QX
D3103
D3104
C14ce55_LED_K
D3105
D109
APHHS1005QWF/D
U3103A
OPA4196IDR
U3103B
LED_PACK_OPA4196IDR_0_IO_UNUSED2
U3103C
LED_PACK_OPA4196IDR_0_IO_UNUSED3
U3103D
LED_PACK_OPA4196IDR_0_IO_UNUSED4
U3103E
C3108
100 nF
C3109
100 nF
D3107
BAT54S,215
COM

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Size: A0
Date:
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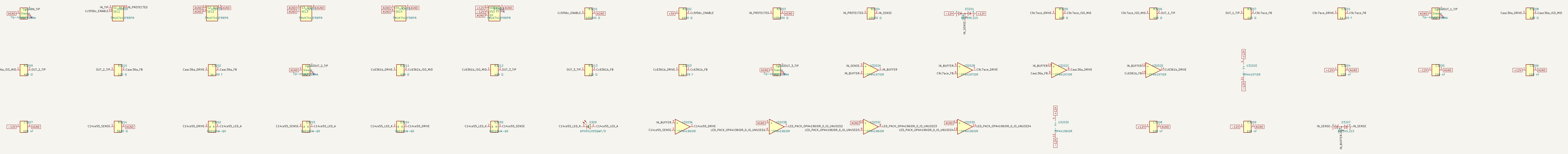
J309
Tip
Sleeve
Tip-normal
WQP518MA
AGND
IN_TIP
U3201A
TMUX7412FRRPR
IN_TIP
Cc5f06c_ENABLE
IN_PROTECTED
U3201B
TMUX7412FRRPR
AGND
U3201C
TMUX7412FRRPR
AGND
U3201D
TMUX7412FRRPR
AGND
U3201E
TMUX7412FRRPR
+12V
-12V
AGND
R3201
100000 Ω
Cc5f06c_ENABLE
AGND
R3202
1000 Ω
+5V
Cc5f06c_ENABLE
R3203
100000 Ω
IN_PROTECTED
AGND
R3204
10000 Ω
IN_PROTECTED
IN_SENSE
D3201
BAV199,215
-12V
+12V
IN_SENSE
R3205
499 Ω
C9c7ace_DRIVE
C9c7ace_ISO_MID
R3206
499 Ω
C9c7ace_ISO_MID
OUT_1_TIP
R3207
100 Ω
OUT_1_TIP
C9c7ace_FB
C3201
1e-09 F
C9c7ace_DRIVE
C9c7ace_FB
J409
Tip-normal
WQP518MA
AGND
OUT_1_TIP
R3208
499 Ω
Caac36a_DRIVE
Caac36a_ISO_MID
R3209
499 Ω
Caac36a_ISO_MID
OUT_2_TIP
R3210
100 Ω
OUT_2_TIP
Caac36a_FB
C3202
1e-09 F
Caac36a_DRIVE
Caac36a_FB
J310
Tip-normal
WQP518MA
AGND
OUT_2_TIP
R3211
499 Ω
Cc6361b_DRIVE
Cc6361b_ISO_MID
R3212
499 Ω
Cc6361b_ISO_MID
OUT_3_TIP
R3213
100 Ω
OUT_3_TIP
Cc6361b_FB
C3203
1e-09 F
Cc6361b_DRIVE
Cc6361b_FB
J410
Tip-normal
WQP518MA
AGND
OUT_3_TIP
U3202A
OPA4197IDR
IN_SENSE
IN_BUFFER
IN_BUFFER
U3202B
OPA4197IDR
IN_BUFFER
C9c7ace_FB
C9c7ace_DRIVE
U3202C
OPA4197IDR
IN_BUFFER
Caac36a_FB
Caac36a_DRIVE
U3202D
OPA4197IDR
IN_BUFFER
Cc6361b_FB
Cc6361b_DRIVE
U3202E
OPA4197IDR
+12V
-12V
C3204
100 nF
+12V
AGND
C3205
100 nF
-12V
AGND
C3206
100 nF
+12V
AGND
C3207
100 nF
-12V
AGND
R3214
5600 Ω
C14ce55_SENSE
AGND
D3202
BAS16GW-QX
C14ce55_DRIVE
C14ce55_LED_A
D3203
BAS16GW-QX
C14ce55_SENSE
C14ce55_LED_A
D3204
BAS16GW-QX
C14ce55_LED_K
C14ce55_DRIVE
D3205
BAS16GW-QX
C14ce55_LED_K
C14ce55_SENSE
D309
APHHS1005QWF/D
C14ce55_LED_K
C14ce55_LED_A
U3203A
OPA4196IDR
IN_BUFFER
C14ce55_SENSE
C14ce55_DRIVE
U3203B
OPA4196IDR
AGND
LED_PACK_OPA4196IDR_0_IO_UNUSED2
LED_PACK_OPA4196IDR_0_IO_UNUSED2
U3203C
OPA4196IDR
AGND
LED_PACK_OPA4196IDR_0_IO_UNUSED3
LED_PACK_OPA4196IDR_0_IO_UNUSED3
U3203D
OPA4196IDR
AGND
LED_PACK_OPA4196IDR_0_IO_UNUSED4
LED_PACK_OPA4196IDR_0_IO_UNUSED4
U3203E
OPA4196IDR
+12V
-12V
C3208
100 nF
+12V
AGND
C3209
100 nF
-12V
AGND
D3207
BAT54S,215
IN_SENSE
IN_SENSE
IN_BUFFER

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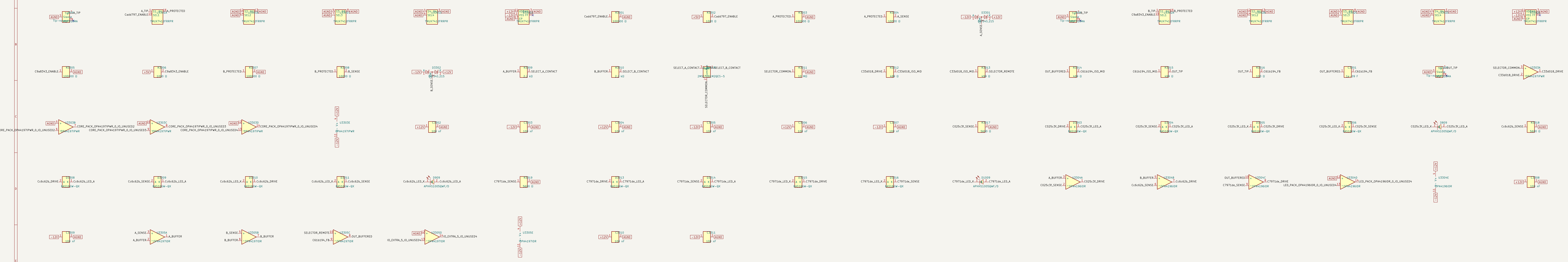

J809 Tip Sleeve Tip-normal TN AGND A_TIP WQP518MA
U3301A TMUX7412FRRPR S1 D1 SEL1 A_TIP Cadd797_ENABLE A_PROTECTED
U3301B TMUX7412FRRPR S2 D2 SEL2 AGND
U3301C TMUX7412FRRPR S3 D3 SEL3 AGND
U3301D TMUX7412FRRPR S4 D4 SEL4 AGND
U3301E TMUX7412FRRPR VDD GND VSS FF EP +12V −12V AGND
R3301 100000 Ω Cadd797_ENABLE AGND
R3302 1000 Ω +5V Cadd797_ENABLE
R3303 100000 Ω A_PROTECTED AGND
R3304 10000 Ω A_PROTECTED A_SENSE
D3301 BAT54S,215 −12V +12V A_SENSE
J909 Tip Sleeve Tip-normal TN AGND B_TIP WQP518MA
U3302A TMUX7412FRRPR S1 D1 SEL1 B_TIP C9a8343_ENABLE B_PROTECTED
U3302B TMUX7412FRRPR S2 D2 SEL2 AGND
U3302C TMUX7412FRRPR S3 D3 SEL3 AGND
U3302D TMUX7412FRRPR S4 D4 SEL4 AGND
U3302E TMUX7412FRRPR VDD GND VSS FF EP +12V −12V AGND
R3305 100000 Ω C9a8343_ENABLE AGND
R3306 1000 Ω +5V C9a8343_ENABLE
R3307 100000 Ω B_PROTECTED AGND
R3308 10000 Ω B_PROTECTED B_SENSE
D3302 BAT54S,215 −12V +12V B_SENSE
R3309 2.2 kΩ A_BUFFER SELECT_A_CONTACT
R3310 2.2 kΩ B_BUFFER SELECT_B_CONTACT
2MS1T1B1M2QES-5 SELECT_A_CONTACT SELECT_B_CONTACT SELECTOR_COMMON
R3311 10 MΩ SELECTOR_COMMON AGND
R3312 499 Ω C33d018_DRIVE C33d018_ISO_MID
R3313 499 Ω C33d018_ISO_MID SELECTOR_REMOTE
R3314 499 Ω OUT_BUFFERED C61b194_ISO_MID
R3315 499 Ω C61b194_ISO_MID OUT_TIP
R3316 100 Ω OUT_TIP C61b194_FB
C3301 1e-09 F OUT_BUFFERED C61b194_FB
J1009 Tip Sleeve Tip-normal TN AGND OUT_TIP WQP518MA
U3303A OPA4197IPWR SELECTOR_COMMON C33d018_DRIVE C33d018_DRIVE
U3303B OPA4197IPWR AGND CORE_PACK_OPA4197IPWR_0_IO_UNUSED2
U3303C OPA4197IPWR AGND CORE_PACK_OPA4197IPWR_0_IO_UNUSED3
U3303D OPA4197IPWR AGND CORE_PACK_OPA4197IPWR_0_IO_UNUSED4
U3303E OPA4197IPWR V+ V− +12V −12V
C3302 100 nF +12V AGND
C3303 100 nF −12V AGND
C3304 100 nF +12V AGND
C3305 100 nF −12V AGND
C3306 100 nF +12V AGND
C3307 100 nF −12V AGND
R3317 5600 Ω C025c3f_SENSE AGND
D3303 BAS16GW-QX C025c3f_DRIVE C025c3f_LED_A
D3304 BAS16GW-QX C025c3f_SENSE C025c3f_LED_A
D3305 BAS16GW-QX C025c3f_LED_K C025c3f_DRIVE
D3306 BAS16GW-QX C025c3f_LED_K C025c3f_SENSE
D809 APHHS1005QWF/D C025c3f_LED_K C025c3f_LED_A
R3318 5600 Ω Cc6c62b_SENSE AGND
D3308 BAS16GW-QX Cc6c62b_DRIVE Cc6c62b_LED_A
D3309 BAS16GW-QX Cc6c62b_SENSE Cc6c62b_LED_A
D3310 BAS16GW-QX Cc6c62b_LED_K Cc6c62b_DRIVE
D3311 BAS16GW-QX Cc6c62b_LED_K Cc6c62b_SENSE
D909 APHHS1005QWF/D Cc6c62b_LED_K Cc6c62b_LED_A
R3319 5600 Ω C7971de_SENSE AGND
D3313 BAS16GW-QX C7971de_DRIVE C7971de_LED_A
D3314 BAS16GW-QX C7971de_SENSE C7971de_LED_A
D3315 BAS16GW-QX C7971de_LED_K C7971de_DRIVE
D3316 BAS16GW-QX C7971de_LED_K C7971de_SENSE
D1009 APHHS1005QWF/D C7971de_LED_K C7971de_LED_A
U3304A OPA4196IDR A_BUFFER C025c3f_SENSE C025c3f_DRIVE
U3304B OPA4196IDR B_BUFFER Cc6c62b_SENSE Cc6c62b_DRIVE
U3304C OPA4196IDR OUT_BUFFERED C7971de_SENSE C7971de_DRIVE
U3304D OPA4196IDR AGND LED_PACK_OPA4196IDR_0_IO_UNUSED4
U3304E OPA4196IDR V+ V− +12V −12V
C3308 100 nF +12V AGND
C3309 100 nF −12V AGND
U3305A OPA4197IDR A_SENSE A_BUFFER A_BUFFER
U3305B OPA4197IDR B_SENSE B_BUFFER B_BUFFER
U3305C OPA4197IDR SELECTOR_REMOTE C61b194_FB OUT_BUFFERED
U3305D OPA4197IDR AGND IO_EXTRA_5_IO_UNUSED4
U3305E OPA4197IDR V+ V− +12V −12V
C3310 100 nF +12V AGND
C3311 100 nF −12V AGND

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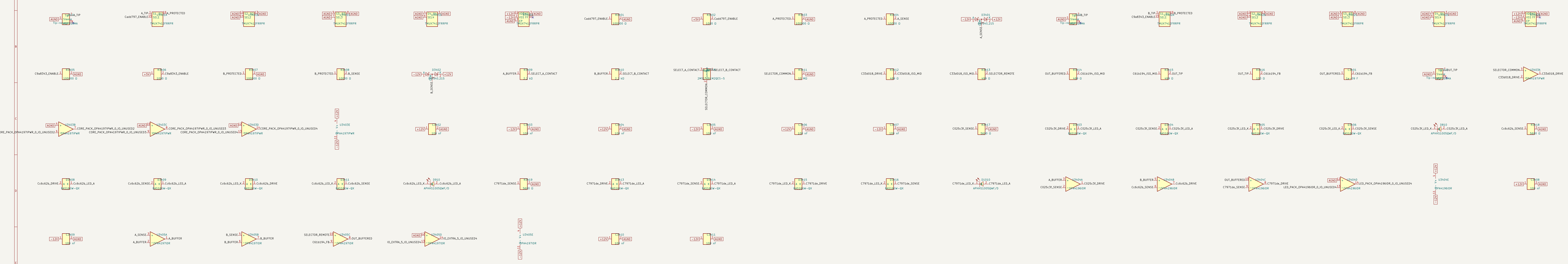

AGND
A_TIP
Tip
Sleeve
Tip-normal
J3401
A_PROTECTED
Cadd797_ENABLE
S1
SEL1
U3401A
TMUX7412FRRPR
S2
SEL2
U3401B
S3
SEL3
U3401C
S4
SEL4
U3401D
+12V
-12V
VDD
GND
VSS
EP
U3401E
R3401
100000 Ω
R3402
+5V
1000 Ω
R3403
R3404
A_SENSE
10000 Ω
D3401
BAT54S,215
B_TIP
J3402
B_PROTECTED
C9a8343_ENABLE
U3402A
U3402B
U3402C
U3402D
U3402E
R3405
R3406
R3407
R3408
B_SENSE
D3402
R3409
A_BUFFER
SELECT_A_CONTACT
2.2 kΩ
R3410
B_BUFFER
SELECT_B_CONTACT
SELECT_A_CONTACT
SELECT_B_CONTACT
SELECTOR_COMMON
2MS1T1B1M2QES-5
R3411
10 MΩ
R3412
C33d018_DRIVE
C33d018_ISO_MID
499 Ω
R3413
SELECTOR_REMOTE
R3414
OUT_BUFFERED
C61b194_ISO_MID
R3415
OUT_TIP
R3416
C61b194_FB
100 Ω
C3401
1e-09 F
J3403
U3403A
OPA4197IPWR
U3403B
CORE_PACK_OPA4197IPWR_0_IO_UNUSED2
U3403C
CORE_PACK_OPA4197IPWR_0_IO_UNUSED3
U3403D
CORE_PACK_OPA4197IPWR_0_IO_UNUSED4
U3403E
V+
V-
C3402
100 nF
C3403
C3404
C3405
C3406
C3407
R3417
C025c3f_SENSE
5600 Ω
D3403
C025c3f_DRIVE
C025c3f_LED_A
BAS16GW-QX
D3404
D3405
C025c3f_LED_K
D3406
D810
APHHS1005QWF/D
R3418
Cc6c62b_SENSE
D3408
Cc6c62b_DRIVE
Cc6c62b_LED_A
D3409
D3410
Cc6c62b_LED_K
D3411
D910
R3419
C7971de_SENSE
D3413
C7971de_DRIVE
C7971de_LED_A
D3414
D3415
C7971de_LED_K
D3416
D1010
U3404A
OPA4196IDR
U3404B
U3404C
U3404D
LED_PACK_OPA4196IDR_0_IO_UNUSED4
U3404E
C3408
C3409
U3405A
OPA4197IDR
U3405B
U3405C
U3405D
IO_EXTRA_5_IO_UNUSED4
U3405E
C3410
C3411

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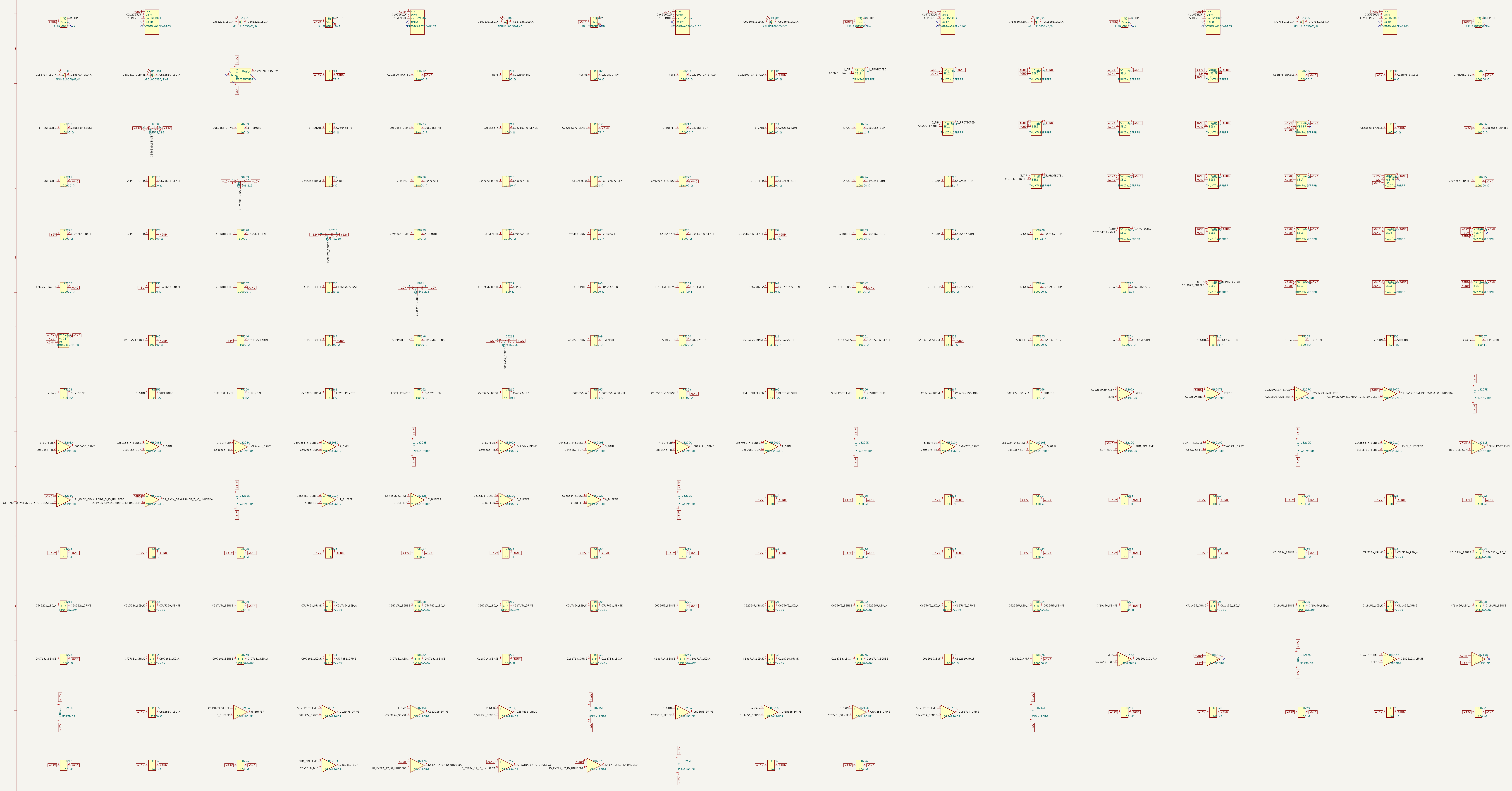

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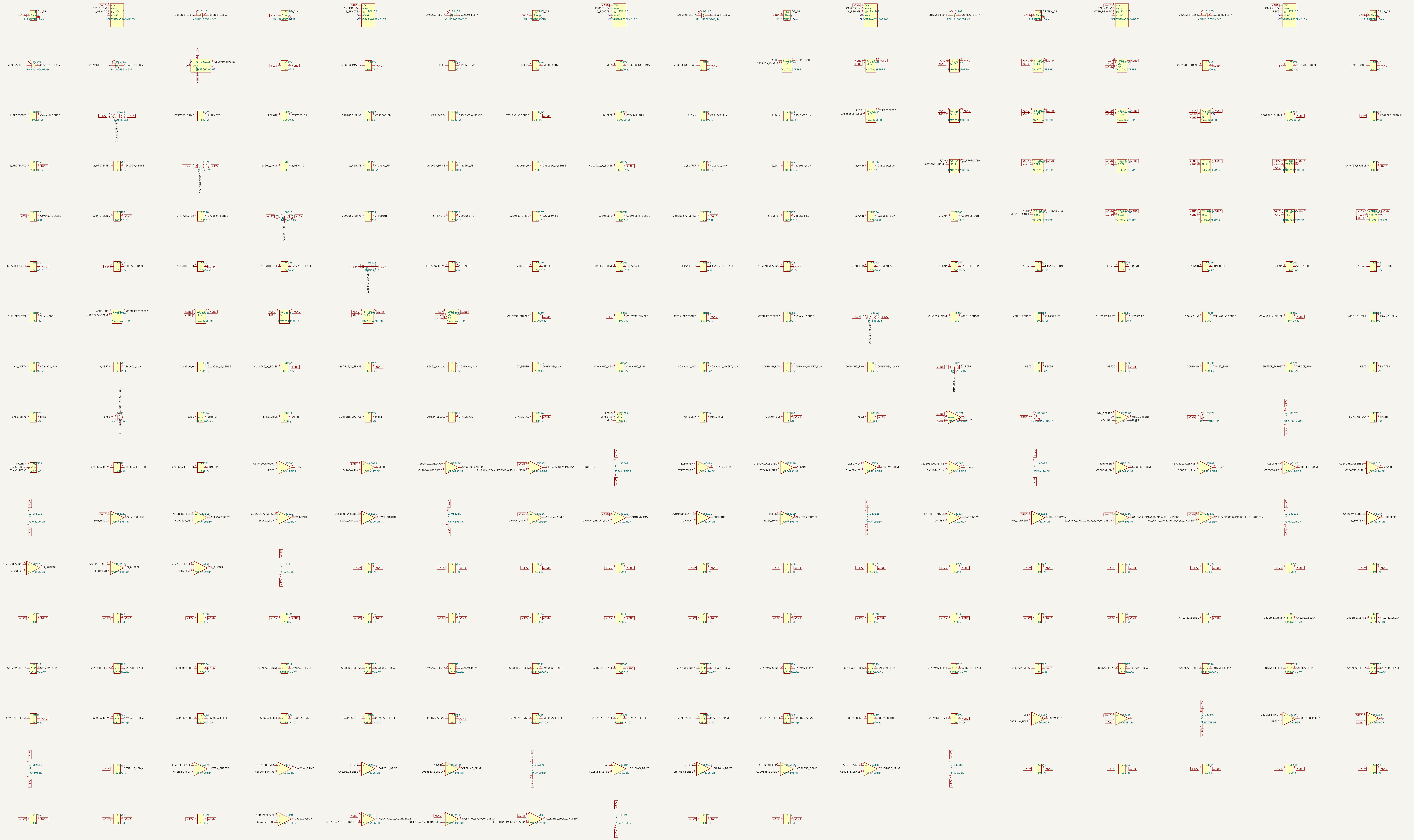
KiCad E.D.A. 10.0.6
Id: 26/36



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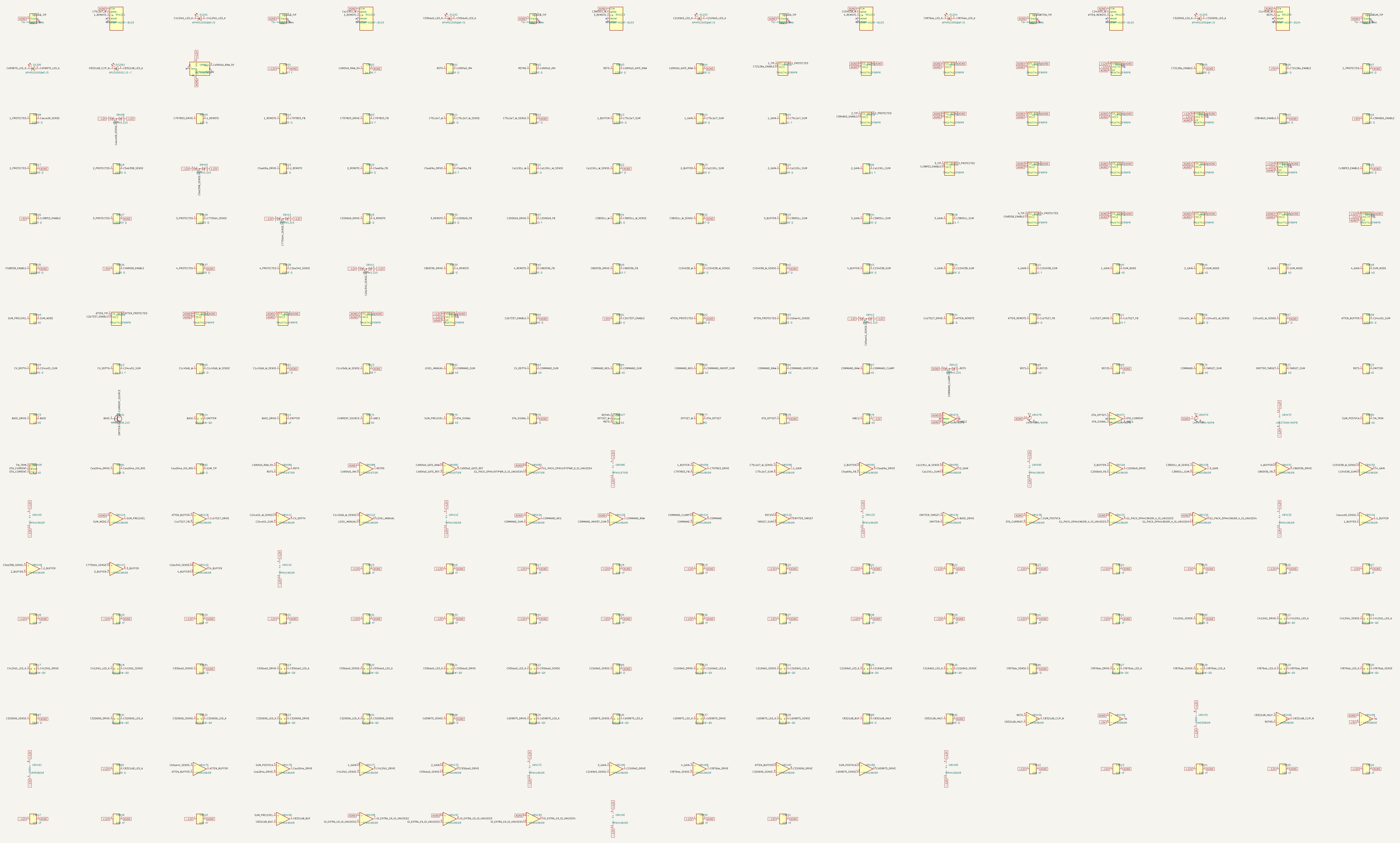
**Rev:**
KiCad E.D.A. 10.0.6
Id: 27/36



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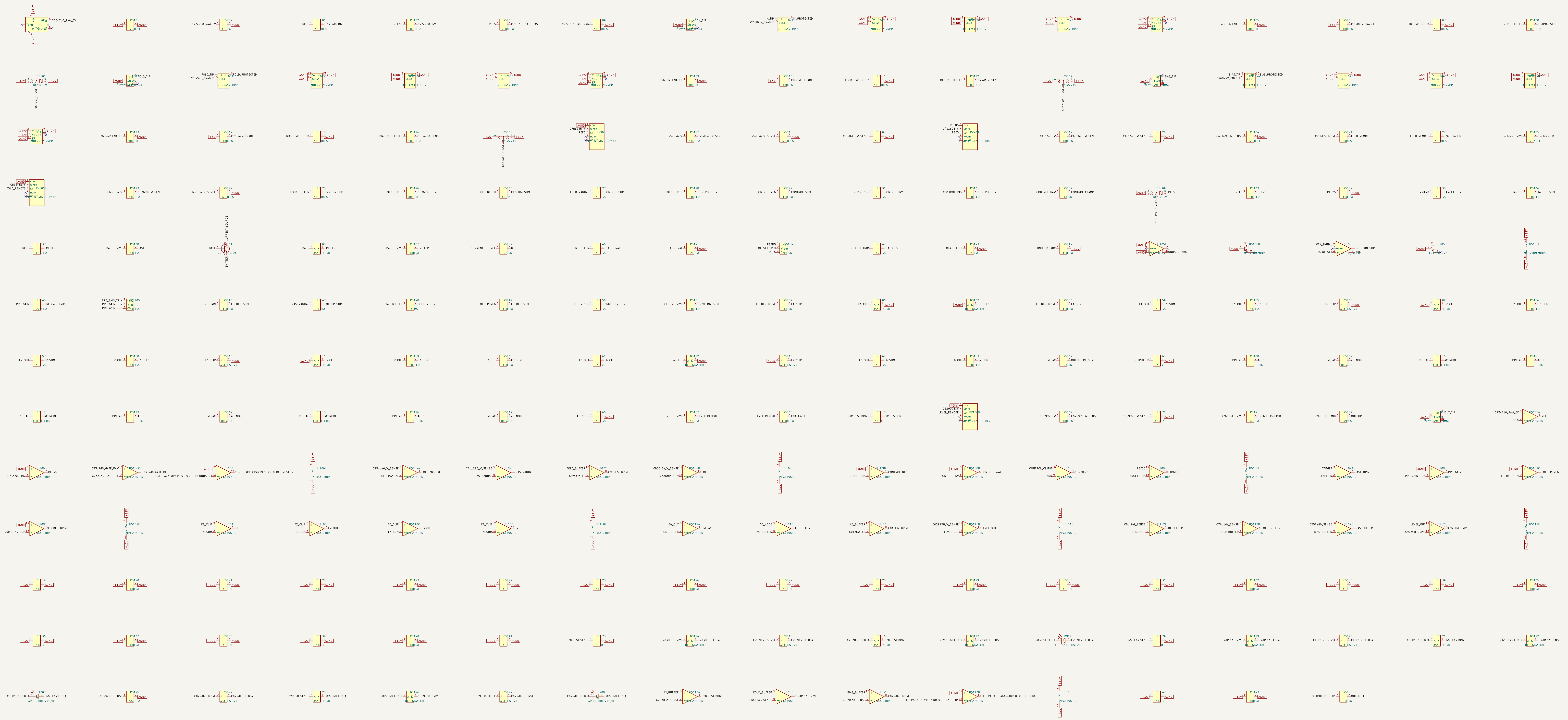
| KiCad E.D.A. 10.0.6 | | Id: 28/36 |



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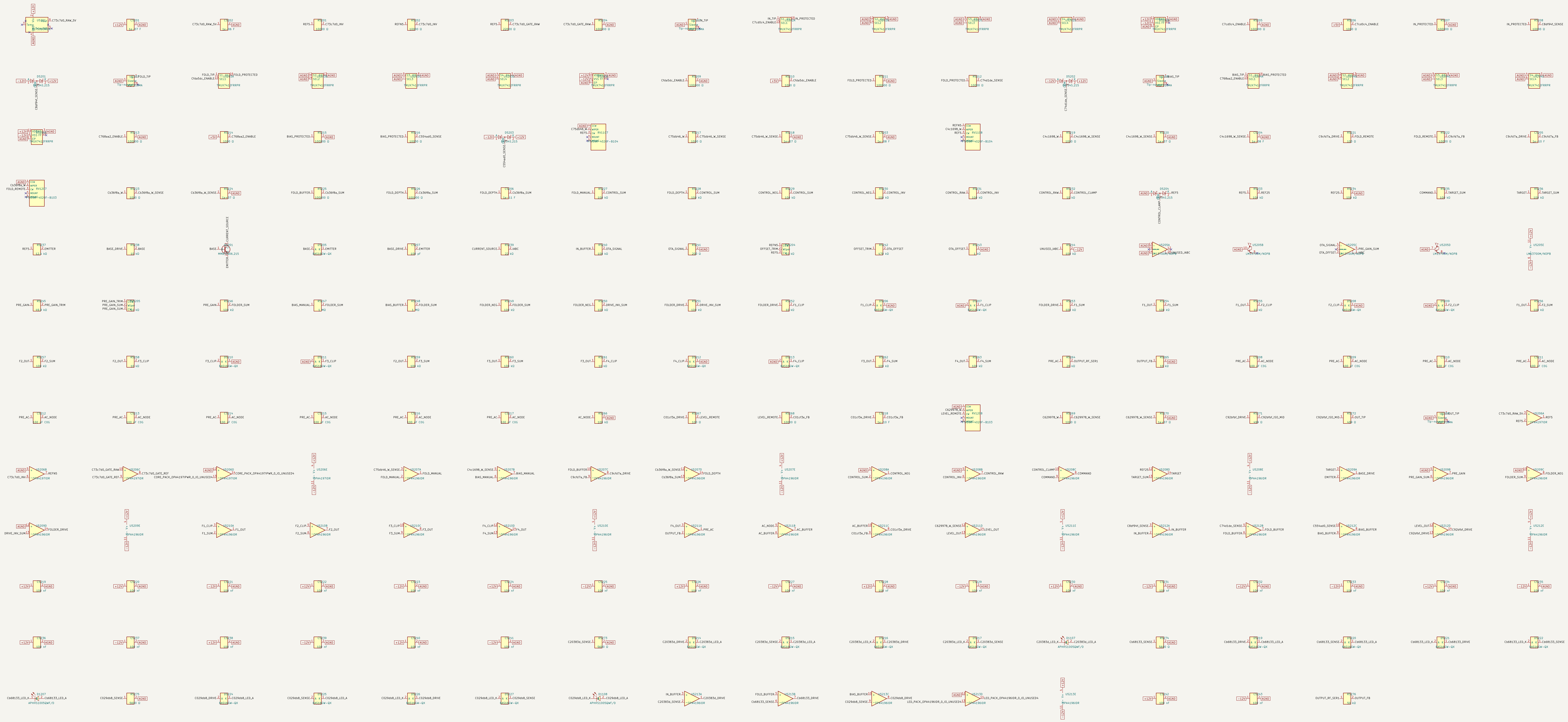
| KiCad E.D.A. 10.0.6 | | Id: 29/36 |



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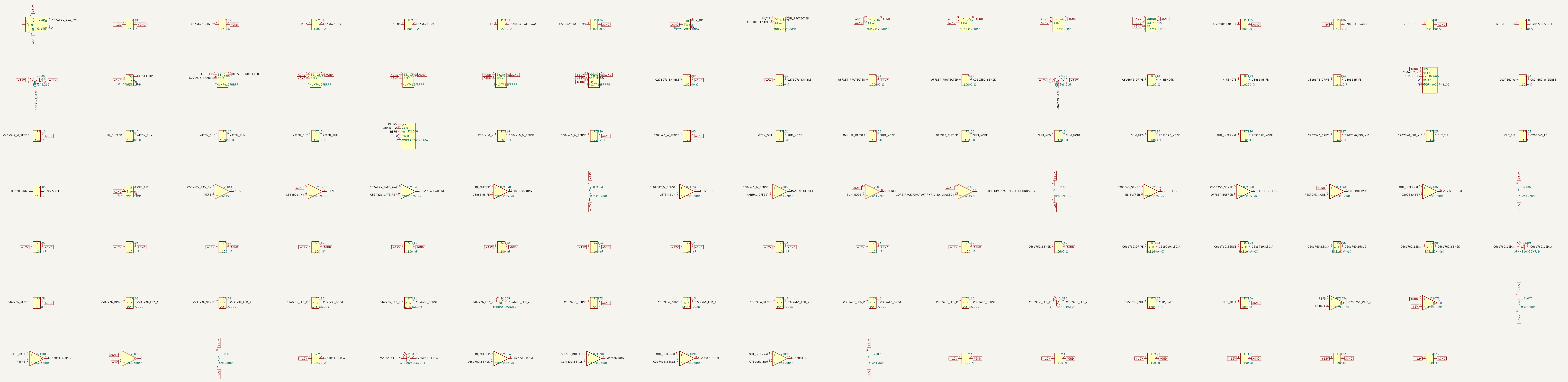

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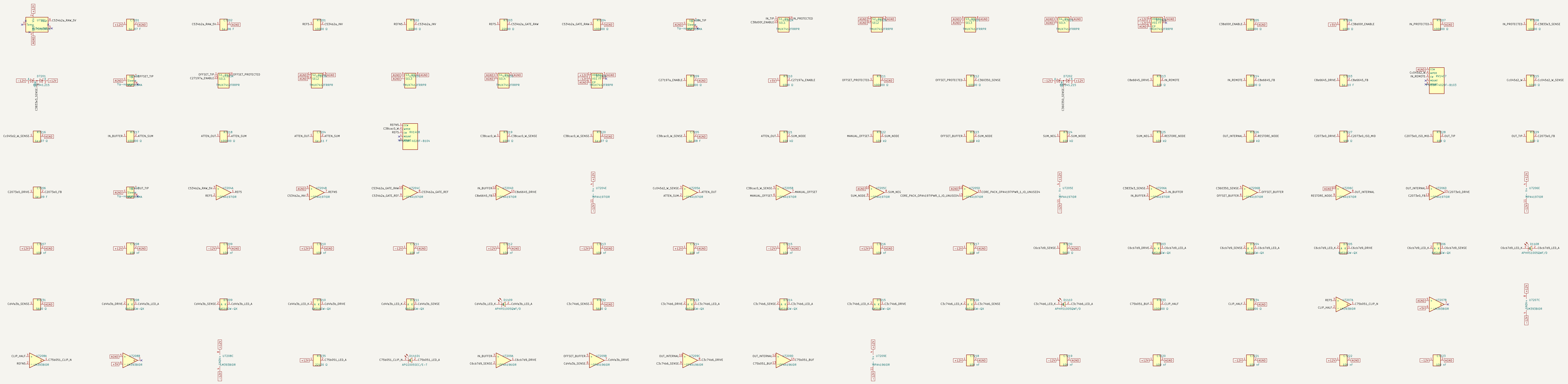

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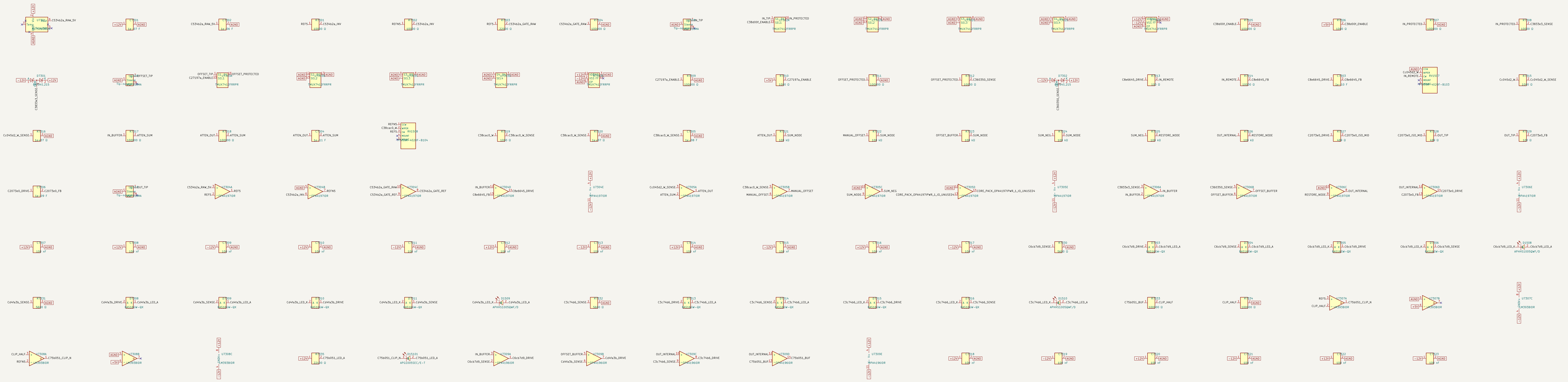
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Vin
Vout
Temp
GND
Trim/NR
REF5050AIDR
+12V
AGND
C534b2a_RAW_5V
C7301
1e-07 F
C7302
1e-06 F
D7301
BAT54S,215
-12V
C3833e3_SENSE
J1509
Tip
Sleeve
OFFSET_TIP
U7303A
TMUX7412FRRPR
OFFSET_PROTECTED
C27197a_ENABLE
R7316
Cc045d2_W_SENSE
1e+07 Ω
R7317
IN_BUFFER
ATTEN_SUM
100000 Ω
R7318
ATTEN_OUT

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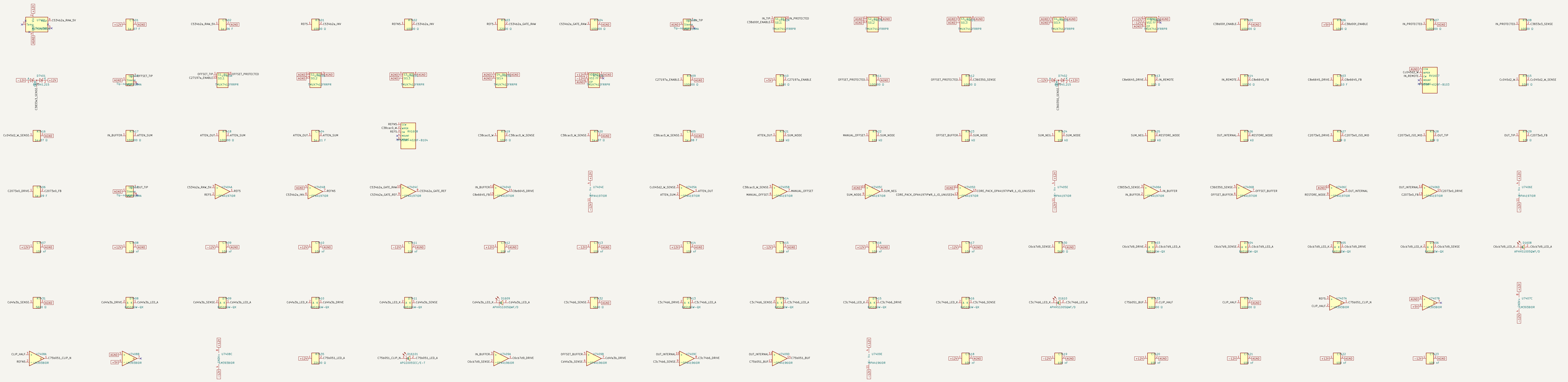
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C7401 +12V AGND 1e-07 F
C7402 C534b2a_RAW_5V AGND 1e-06 F
R7401 REF5 C534b2a_INV 10000 Ω
R7402 REFN5 C534b2a_INV 10000 Ω
R7403 REF5 C534b2a_GATE_RAW 22000 Ω
R7404 C534b2a_GATE_RAW AGND 100000 Ω
J1608 AGND IN_TIP Tip Sleeve Tip-normal
U7402A IN_TIP C38d00f_ENABLE IN_PROTECTED S1 D1 SEL1 TMUX7412FRRPR
U7402B AGND AGND AGND S2 D2 SEL2 TMUX7412FRRPR
U7402C AGND AGND AGND S3 D3 SEL3 TMUX7412FRRPR
U7402D AGND AGND AGND S4 D4 SEL4 TMUX7412FRRPR
U7402E +12V -12V AGND AGND VDD GND VSS FF EP TMUX7412FRRPR
R7405 C38d00f_ENABLE AGND 100000 Ω
R7406 +5V C38d00f_ENABLE 1000 Ω
R7407 IN_PROTECTED AGND 100000 Ω
R7408 IN_PROTECTED C3833e3_SENSE 10000 Ω
D7401 -12V +12V A K COM BAT54S,215 C3833e3_SENSE
J1609 AGND OFFSET_TIP Tip Sleeve Tip-normal
U7403A OFFSET_TIP C27197a_ENABLE OFFSET_PROTECTED S1 D1 SEL1 TMUX7412FRRPR
U7403B AGND AGND AGND S2 D2 SEL2 TMUX7412FRRPR
U7403C AGND AGND AGND S3 D3 SEL3 TMUX7412FRRPR
U7403D AGND AGND AGND S4 D4 SEL4 TMUX7412FRRPR
U7403E +12V -12V AGND AGND VDD GND VSS FF EP TMUX7412FRRPR
R7409 C27197a_ENABLE AGND 100000 Ω
R7410 +5V C27197a_ENABLE 1000 Ω
R7411 OFFSET_PROTECTED AGND 100000 Ω
R7412 OFFSET_PROTECTED C360350_SENSE 10000 Ω
D7402 -12V +12V A K COM BAT54S,215 C360350_SENSE
R7413 C8e6645_DRIVE IN_REMOTE 100 Ω
R7414 IN_REMOTE C8e6645_FB 10000 Ω
C7403 C8e6645_DRIVE C8e6645_FB 1e-10 F
RV1607 AGND Cc045d2_W IN_REMOTE CCW WIPER CW MOUNT MOUNT PTV09A-4020F-B103
R7415 Cc045d2_W Cc045d2_W_SENSE 1000 Ω
R7416 Cc045d2_W_SENSE AGND 1e+07 Ω
R7417 IN_BUFFER ATTEN_SUM 100000 Ω
R7418 ATTEN_OUT ATTEN_SUM 100000 Ω
C7404 ATTEN_OUT ATTEN_SUM 1e-11 F
RV1608 REFN5 C38cac0_W REF5 CCW WIPER CW MOUNT MOUNT PTV09A-4020F-B104
R7419 C38cac0_W C38cac0_W_SENSE 1000 Ω
R7420 C38cac0_W_SENSE AGND 1e+07 Ω
C7405 C38cac0_W_SENSE AGND 1e-08 F
R7421 ATTEN_OUT SUM_NODE 100 kΩ
R7422 MANUAL_OFFSET SUM_NODE 100 kΩ
R7423 OFFSET_BUFFER SUM_NODE 100 kΩ
R7424 SUM_NEG SUM_NODE 100 kΩ
R7425 SUM_NEG RESTORE_NODE 100 kΩ
R7426 OUT_INTERNAL RESTORE_NODE 100 kΩ
R7427 C2073e0_DRIVE C2073e0_ISO_MID 499 Ω
R7428 C2073e0_ISO_MID OUT_TIP 499 Ω
R7429 OUT_TIP C2073e0_FB 100 Ω
C7406 C2073e0_DRIVE C2073e0_FB 1e-09 F
J1610 AGND OUT_TIP Tip Sleeve Tip-normal
U7404A C534b2a_RAW_5V REF5 REF5 OPA4197IDR
U7404B AGND C534b2a_INV REFN5 OPA4197IDR
U7404C C534b2a_GATE_RAW C534b2a_GATE_REF C534b2a_GATE_REF OPA4197IDR
U7404D IN_BUFFER C8e6645_FB C8e6645_DRIVE OPA4197IDR
U7404E +12V -12V V+ V- OPA4197IDR
U7405A Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT OPA4197IDR
U7405B C38cac0_W_SENSE MANUAL_OFFSET MANUAL_OFFSET OPA4197IDR
U7405C AGND SUM_NODE SUM_NEG OPA4197IDR
U7405D AGND CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 OPA4197IDR
U7405E +12V -12V V+ V- OPA4197IDR
U7406A C3833e3_SENSE IN_BUFFER IN_BUFFER OPA4197IDR
U7406B C360350_SENSE OFFSET_BUFFER OFFSET_BUFFER OPA4197IDR
U7406C AGND RESTORE_NODE OUT_INTERNAL OPA4197IDR
U7406D OUT_INTERNAL C2073e0_FB C2073e0_DRIVE OPA4197IDR
U7406E +12V -12V V+ V- OPA4197IDR
C7407 +12V AGND 100 nF
C7408 +12V AGND 100 nF
C7409 -12V AGND 100 nF
C7410 +12V AGND 100 nF
C7411 -12V AGND 100 nF
C7412 +12V AGND 100 nF
C7413 -12V AGND 100 nF
C7414 +12V AGND 100 nF
C7415 -12V AGND 100 nF
C7416 +12V AGND 100 nF
C7417 -12V AGND 100 nF
R7430 C6cb7d9_SENSE AGND 5600 Ω
D7403 C6cb7d9_DRIVE C6cb7d9_LED_A A K BAS16GW-QX
D7404 C6cb7d9_SENSE C6cb7d9_LED_A A K BAS16GW-QX
D7405 C6cb7d9_LED_K C6cb7d9_DRIVE A K BAS16GW-QX
D7406 C6cb7d9_LED_K C6cb7d9_SENSE A K BAS16GW-QX
D1608 C6cb7d9_LED_K C6cb7d9_LED_A APHHS1005QWF/D
R7431 Cd4fa3b_SENSE AGND 5600 Ω
D7408 Cd4fa3b_DRIVE Cd4fa3b_LED_A A K BAS16GW-QX
D7409 Cd4fa3b_SENSE Cd4fa3b_LED_A A K BAS16GW-QX
D7410 Cd4fa3b_LED_K Cd4fa3b_DRIVE A K BAS16GW-QX
D7411 Cd4fa3b_LED_K Cd4fa3b_SENSE A K BAS16GW-QX
D1609 Cd4fa3b_LED_K Cd4fa3b_LED_A APHHS1005QWF/D
R7432 C3c74b6_SENSE AGND 5600 Ω
D7413 C3c74b6_DRIVE C3c74b6_LED_A A K BAS16GW-QX
D7414 C3c74b6_SENSE C3c74b6_LED_A A K BAS16GW-QX
D7415 C3c74b6_LED_K C3c74b6_DRIVE A K BAS16GW-QX
D7416 C3c74b6_LED_K C3c74b6_SENSE A K BAS16GW-QX
D1610 C3c74b6_LED_K C3c74b6_LED_A APHHS1005QWF/D
R7433 C75b051_BUF CLIP_HALF 100000 Ω
R7434 CLIP_HALF AGND 100000 Ω
U7407A REF5 CLIP_HALF C75b051_CLIP_N LM393BIDR
U7407B AGND +5V LM393BIDR
U7407C +12V -12V V+ GND LM393BIDR
U7408A CLIP_HALF REFN5 C75b051_CLIP_N LM393BIDR
U7408B AGND +5V LM393BIDR
U7408C +12V -12V V+ GND LM393BIDR
R7435 +12V C75b051_LED_A 22100 Ω
D16101 C75b051_CLIP_N C75b051_LED_A APG1005SEC/E-T
U7409A IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE OPA4196IDR
U7409B OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE OPA4196IDR
U7409C OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE OPA4196IDR
U7409D OUT_INTERNAL C75b051_BUF C75b051_BUF OPA4196IDR
U7409E +12V -12V V+ V- OPA4196IDR
C7418 +12V AGND 100 nF
C7419 -12V AGND 100 nF
C7420 +12V AGND 100 nF
C7421 -12V AGND 100 nF
C7422 +12V AGND 100 nF
C7423 -12V AGND 100 nF

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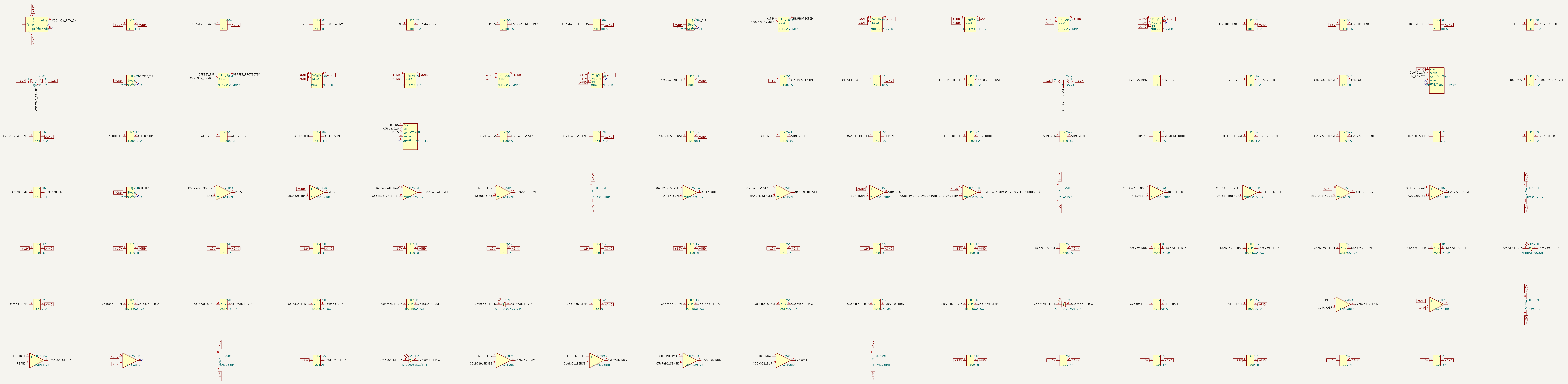

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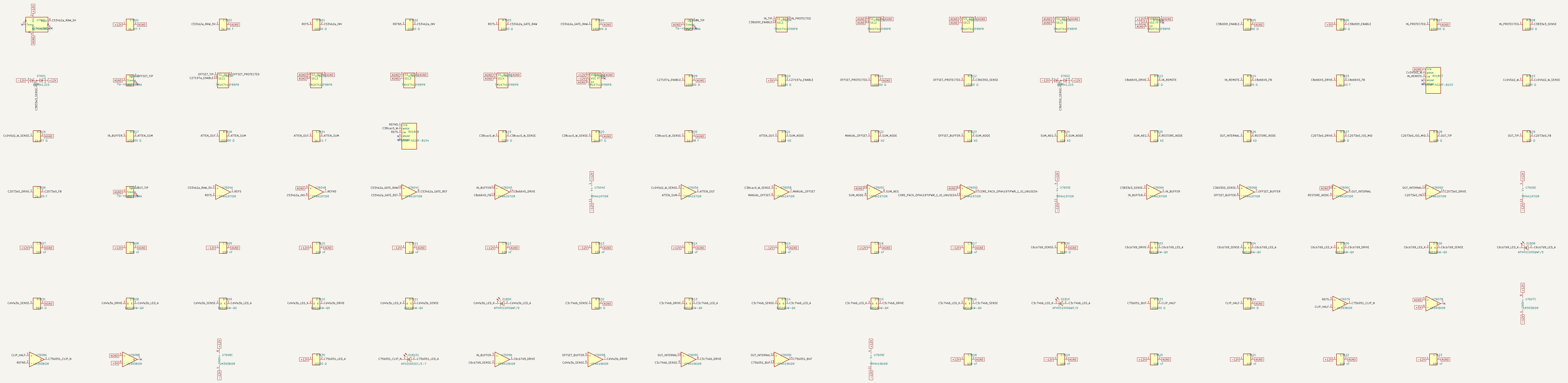
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C7601 1e-07 F +12V AGND
C7602 1e-06 F C534b2a_RAW_5V AGND
R7601 10000 Ω REF5 C534b2a_INV
R7602 10000 Ω REFN5 C534b2a_INV
R7603 22000 Ω REF5 C534b2a_GATE_RAW
R7604 100000 Ω C534b2a_GATE_RAW AGND
Tip Sleeve Tip-normal AGND IN_TIP
U7602A TMUX7412FRRPR S1 D1 SEL1 IN_TIP C38d00f_ENABLE IN_PROTECTED
U7602B TMUX7412FRRPR S2 D2 SEL2 AGND
U7602C TMUX7412FRRPR S3 D3 SEL3 AGND
U7602D TMUX7412FRRPR S4 D4 SEL4 AGND
U7602E TMUX7412FRRPR VDD GND VSS FF EP +12V −12V AGND
R7605 100000 Ω C38d00f_ENABLE AGND
R7606 1000 Ω +5V C38d00f_ENABLE
R7607 100000 Ω IN_PROTECTED AGND
R7608 10000 Ω IN_PROTECTED C3833e3_SENSE
D7601 BAT54S,215 −12V +12V A K COM C3833e3_SENSE
Tip Sleeve Tip-normal AGND OFFSET_TIP
U7603A TMUX7412FRRPR S1 D1 SEL1 OFFSET_TIP C27197a_ENABLE OFFSET_PROTECTED
U7603B TMUX7412FRRPR S2 D2 SEL2 AGND
U7603C TMUX7412FRRPR S3 D3 SEL3 AGND
U7603D TMUX7412FRRPR S4 D4 SEL4 AGND
U7603E TMUX7412FRRPR VDD GND VSS FF EP +12V −12V AGND
R7609 100000 Ω C27197a_ENABLE AGND
R7610 1000 Ω +5V C27197a_ENABLE
R7611 100000 Ω OFFSET_PROTECTED AGND
R7612 10000 Ω OFFSET_PROTECTED C360350_SENSE
D7602 BAT54S,215 −12V +12V A K COM C360350_SENSE
R7613 100 Ω C8e6645_DRIVE IN_REMOTE
R7614 10000 Ω IN_REMOTE C8e6645_FB
C7603 1e-10 F C8e6645_DRIVE C8e6645_FB
RV1807 PTV09A-4020F-B103 CCW WIPER CW MOUNT AGND Cc045d2_W IN_REMOTE
R7615 1000 Ω Cc045d2_W Cc045d2_W_SENSE
R7616 1e+07 Ω Cc045d2_W_SENSE AGND
R7617 100000 Ω IN_BUFFER ATTEN_SUM
R7618 100000 Ω ATTEN_OUT ATTEN_SUM
C7604 1e-11 F ATTEN_OUT ATTEN_SUM
RV1808 PTV09A-4020F-B104 CCW WIPER CW MOUNT REFN5 C38cac0_W REF5
R7619 1000 Ω C38cac0_W C38cac0_W_SENSE
R7620 1e+07 Ω C38cac0_W_SENSE AGND
C7605 1e-08 F C38cac0_W_SENSE AGND
R7621 100 kΩ ATTEN_OUT SUM_NODE
R7622 100 kΩ MANUAL_OFFSET SUM_NODE
R7623 100 kΩ OFFSET_BUFFER SUM_NODE
R7624 100 kΩ SUM_NEG SUM_NODE
R7625 100 kΩ SUM_NEG RESTORE_NODE
R7626 100 kΩ OUT_INTERNAL RESTORE_NODE
R7627 499 Ω C2073e0_DRIVE C2073e0_ISO_MID
R7628 499 Ω C2073e0_ISO_MID OUT_TIP
R7629 100 Ω OUT_TIP C2073e0_FB
C7606 1e-09 F C2073e0_DRIVE C2073e0_FB
Tip Sleeve Tip-normal AGND OUT_TIP
U7604A OPA4197IDR C534b2a_RAW_5V REF5 REF5
U7604B OPA4197IDR AGND C534b2a_INV REFN5
U7604C OPA4197IDR C534b2a_GATE_RAW C534b2a_GATE_REF C534b2a_GATE_REF
U7604D OPA4197IDR IN_BUFFER C8e6645_FB C8e6645_DRIVE
U7604E OPA4197IDR V+ V− +12V −12V
U7605A OPA4197IDR Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT
U7605B OPA4197IDR C38cac0_W_SENSE MANUAL_OFFSET MANUAL_OFFSET
U7605C OPA4197IDR AGND SUM_NODE SUM_NEG
U7605D OPA4197IDR AGND CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 CORE_PACK_OPA4197IPWR_1_IO_UNUSED4
U7605E OPA4197IDR V+ V− +12V −12V
U7606A OPA4197IDR C3833e3_SENSE IN_BUFFER IN_BUFFER
U7606B OPA4197IDR C360350_SENSE OFFSET_BUFFER OFFSET_BUFFER
U7606C OPA4197IDR AGND RESTORE_NODE OUT_INTERNAL
U7606D OPA4197IDR OUT_INTERNAL C2073e0_FB C2073e0_DRIVE
U7606E OPA4197IDR V+ V− +12V −12V
C7607 100 nF +12V AGND
C7608 100 nF +12V AGND
C7609 100 nF −12V AGND
C7610 100 nF +12V AGND
C7611 100 nF −12V AGND
C7612 100 nF +12V AGND
C7613 100 nF −12V AGND
C7614 100 nF +12V AGND
C7615 100 nF −12V AGND
C7616 100 nF +12V AGND
C7617 100 nF −12V AGND
R7630 5600 Ω C6cb7d9_SENSE AGND
D7603 BAS16GW-QX A K C6cb7d9_DRIVE C6cb7d9_LED_A
D7604 BAS16GW-QX A K C6cb7d9_SENSE C6cb7d9_LED_A
D7605 BAS16GW-QX A K C6cb7d9_LED_K C6cb7d9_DRIVE
D7606 BAS16GW-QX A K C6cb7d9_LED_K C6cb7d9_SENSE
D1808 APHHS1005QWF/D C6cb7d9_LED_K C6cb7d9_LED_A
R7631 5600 Ω Cd4fa3b_SENSE AGND
D7608 BAS16GW-QX A K Cd4fa3b_DRIVE Cd4fa3b_LED_A
D7609 BAS16GW-QX A K Cd4fa3b_SENSE Cd4fa3b_LED_A
D7610 BAS16GW-QX A K Cd4fa3b_LED_K Cd4fa3b_DRIVE
D7611 BAS16GW-QX A K Cd4fa3b_LED_K Cd4fa3b_SENSE
D1809 APHHS1005QWF/D Cd4fa3b_LED_K Cd4fa3b_LED_A
R7632 5600 Ω C3c74b6_SENSE AGND
D7613 BAS16GW-QX A K C3c74b6_DRIVE C3c74b6_LED_A
D7614 BAS16GW-QX A K C3c74b6_SENSE C3c74b6_LED_A
D7615 BAS16GW-QX A K C3c74b6_LED_K C3c74b6_DRIVE
D7616 BAS16GW-QX A K C3c74b6_LED_K C3c74b6_SENSE
D1810 APHHS1005QWF/D C3c74b6_LED_K C3c74b6_LED_A
R7633 100000 Ω C75b051_BUF CLIP_HALF
R7634 100000 Ω CLIP_HALF AGND
U7607A LM393BIDR REF5 CLIP_HALF C75b051_CLIP_N
U7607B LM393BIDR AGND +5V
U7607C LM393BIDR V+ GND +12V −12V
U7608A LM393BIDR CLIP_HALF REFN5 C75b051_CLIP_N
U7608B LM393BIDR AGND +5V
U7608C LM393BIDR V+ GND +12V −12V
R7635 22100 Ω +12V C75b051_LED_A
D18101 APG1005SEC/E-T C75b051_CLIP_N C75b051_LED_A
U7609A OPA4196IDR IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE
U7609B OPA4196IDR OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE
U7609C OPA4196IDR OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE
U7609D OPA4196IDR OUT_INTERNAL C75b051_BUF C75b051_BUF
U7609E OPA4196IDR V+ V− +12V −12V
C7618 100 nF +12V AGND
C7619 100 nF −12V AGND
C7620 100 nF +12V AGND
C7621 100 nF −12V AGND
C7622 100 nF +12V AGND
C7623 100 nF −12V AGND

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